

Principle of Communication System		
Lab #:	04	
Lab Title:	Continuous Waveform Transmission and Receiving Using USRP	
Submitted by:		
Names		Registration #
Sumaira Safeer		FA22-BCE-019

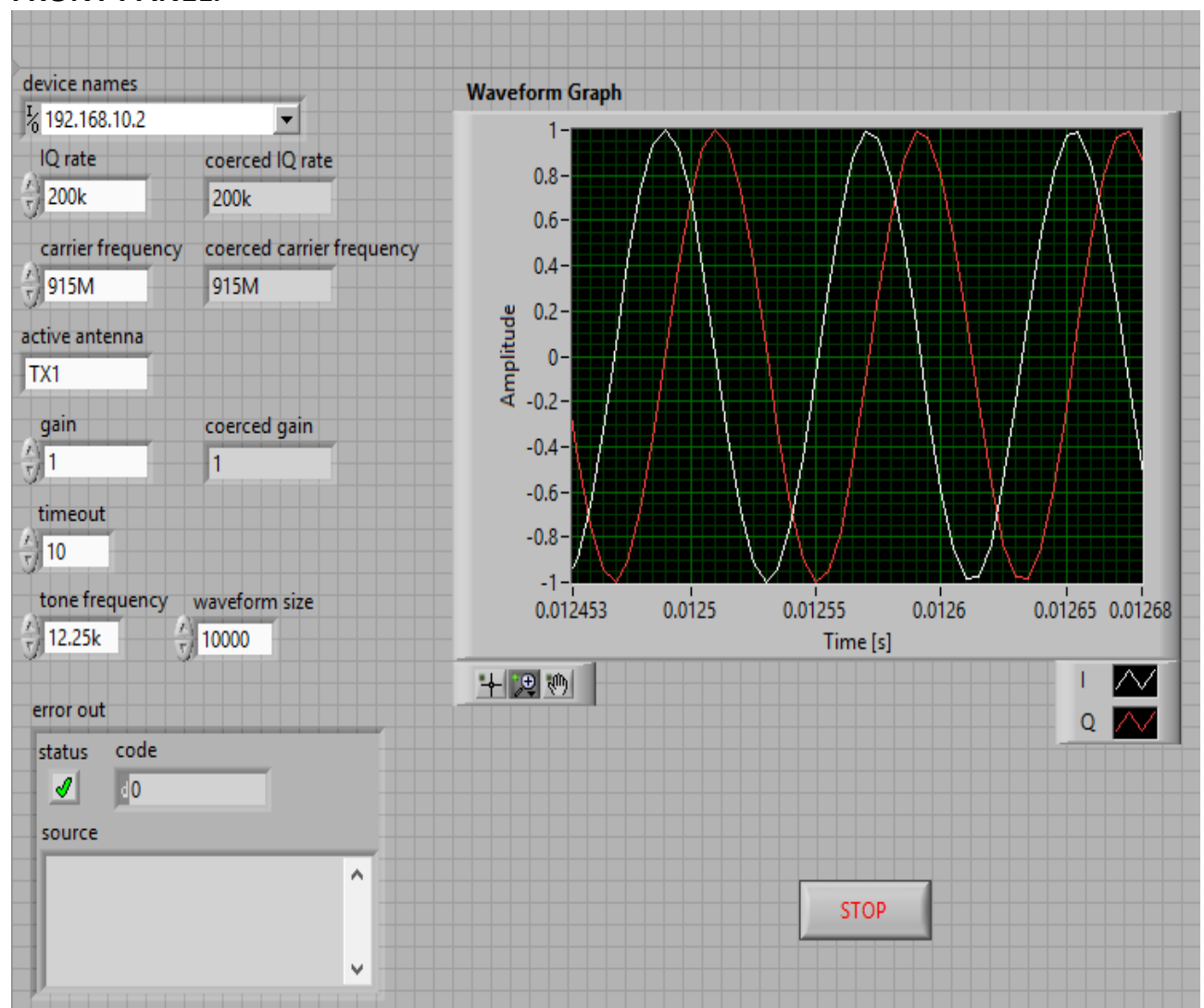
Rubrics name & number					Marks	
					In-Lab	Post-Lab
Engineering Knowledge	R2: Use of Engineering Knowledge and follow Experiment Procedures: Ability to follow experimental procedures, control variables, and record procedural steps on lab report.					
Problem Analysis	R6: Experimental Data Analysis: Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.					
Design	R8: Best Coding Standards: Ability to follow the coding standards and programming practices.					
Modern Tools Usage	R9: Understand Tools: Ability to describe and explain the principles behind and applicability of engineering tools.					
Individual and Teamwork	R12: Individual Work Contributions: Ability to carry out individual responsibilities.					
	R13: Management of Team Work: Ability to appreciate, understand and work with multidisciplinary team members.					
Rubrics #	R2	R6	R8	R9	R12	R13
In Lab						
Post- Lab						

Continuous Waveform Transmission Using USRP (Universal Software Radio Peripheral)

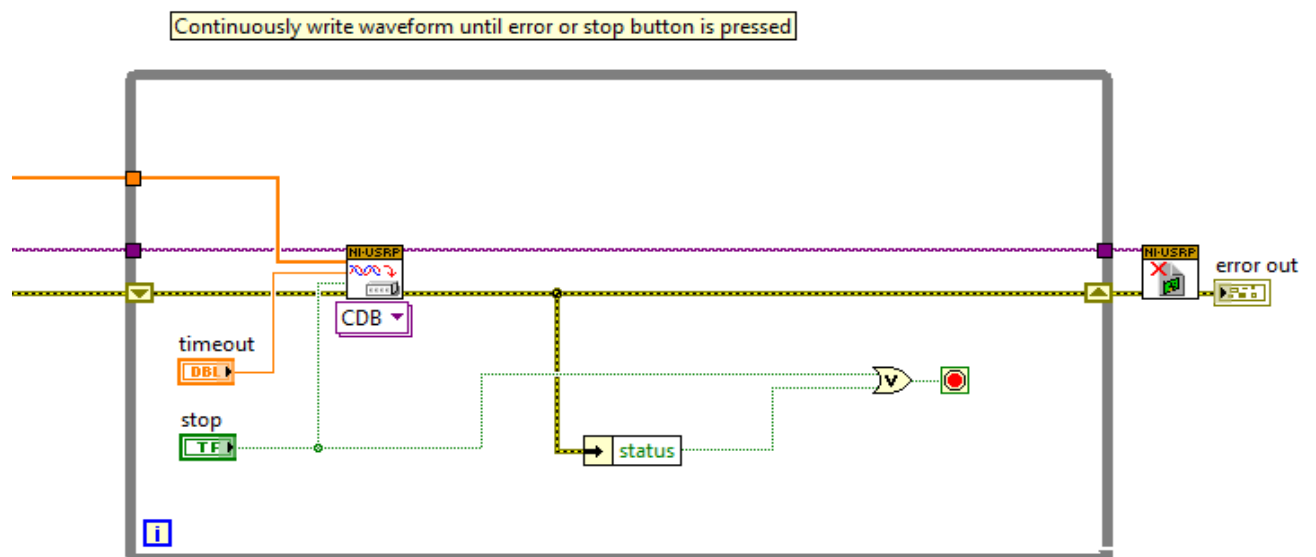
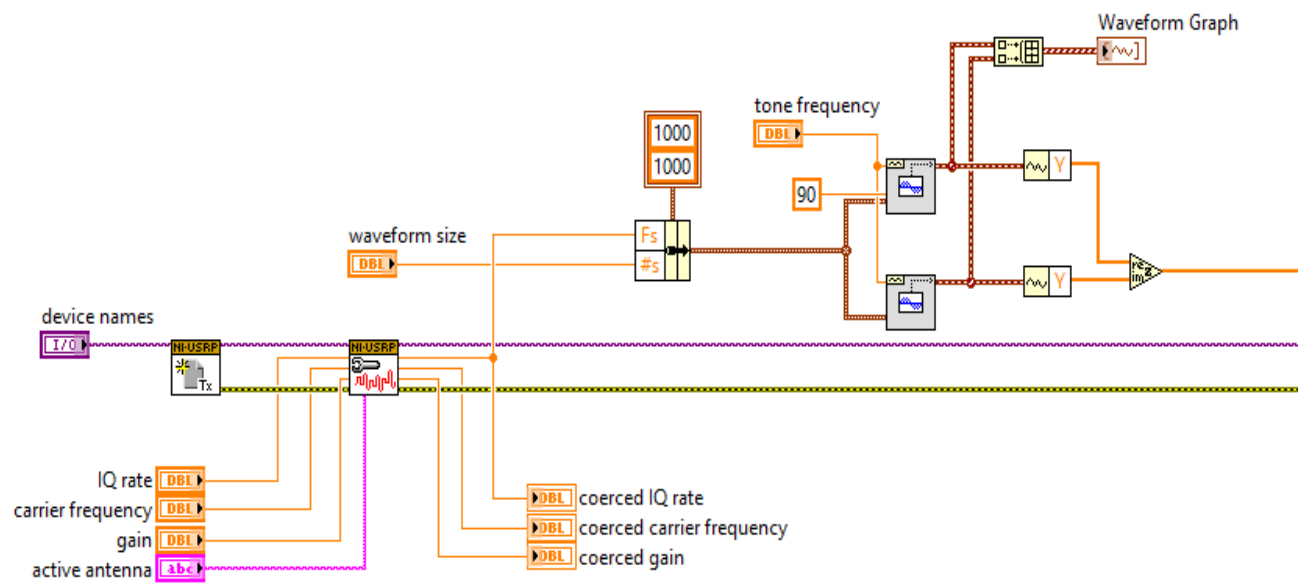
Objective:

- 1) Introduction to USRP and SDR.
- 2) Implementation of Continuous Waveform Transmission Using USRP in LabVIEW.
- 3) Observations of graphical output

FRONT PANEL:



Block Diagram:



Conclusion:

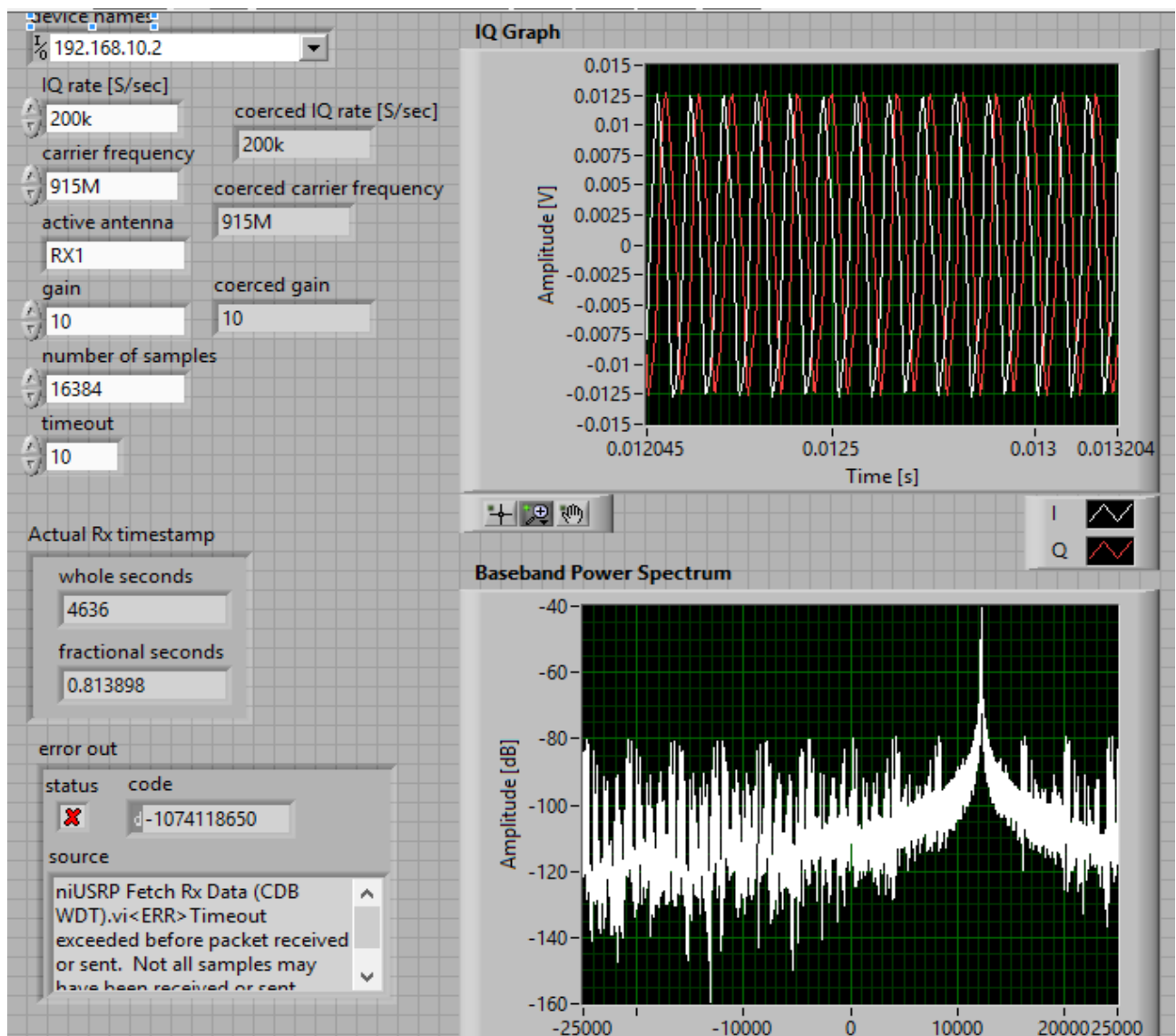
Continuous waveform with IQ Rate of 200K S/Sec with 915MHz carrier frequency was successfully transmitted over TX1/RX1 (active antenna) port.

Continuous Waveform Receiving Using USRP (Universal Software Radio Peripheral)

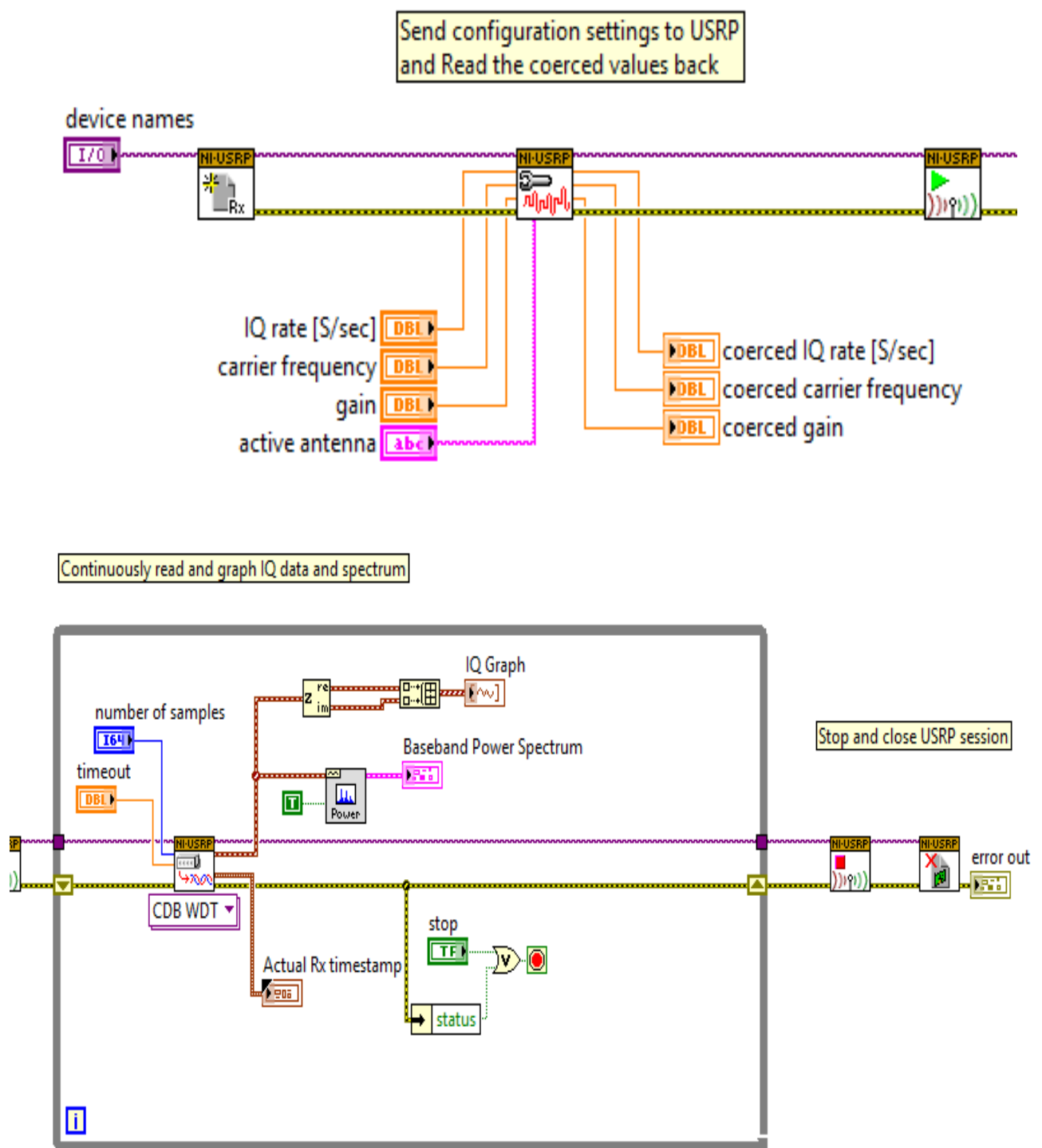
Objective:

- 1) Implementation of Continuous Waveform Receiving Using USRP in LabVIEW.
- 2) Observations of graphical output.

FRONT PANEL:



Block Diagram:



Conclusion:

Continuous waveform with IQ Rate of 200k S/Sec with 915MHz carrier frequency was successfully received over TX1/RX1 (active antenna) port.